

PHASE 6

Project Testing Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To verify that the implemented backend behaves correctly, protected routes are secured, data is processed properly, and AI-powered endpoints return usable results.

6.1 Testing Approach

The backend can be started locally with Node.js after installing dependencies and configuring MongoDB and environment variables. API behavior is then exercised using Postman or Thunder Client.

6.2 Authentication Tests

- Registration is checked to confirm that new users are accepted and stored securely.
- Login is tested to verify valid credentials and JWT token generation.
- Protected endpoints are checked to ensure requests without valid authentication are not permitted.
- Role-restricted operations are tested with appropriate student and administrator accounts.

6.3 Study Material Tests

- Upload requests are checked for successful acceptance of learning content.
- Stored material is verified against the submitted user and content information.
- Invalid or unsupported uploads are checked through the upload validation layer.

6.4 AI Feature Tests

- Summary generation is tested using uploaded material as the source.
- Flashcard generation is checked for question-and-answer output.
- Quiz generation is checked for multiple-choice questions.
- Study-plan generation is tested with learning goals and available study time.

6.5 API Endpoints Used for Verification

- POST /api/auth/register
- POST /api/auth/login
- POST /api/material/upload
- POST /api/materials/:id/summarize
- POST /api/ai/flashcards
- POST /api/ai/quiz
- POST /api/ai/study-plan

6.6 Expected Test Evidence

- Successful HTTP responses for valid requests.
- Correct authentication behavior and token issuance.
- Secure rejection of unauthorized requests.
- Correct persistence of user and learning-resource data.
- Meaningful AI-generated responses for supported operations.

6.7 Testing Result

Testing provides evidence that the major REST APIs, security controls, database interactions, and AI functions operate as intended under the documented usage flow.